

Le Na Hoang

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Detail-oriented Mechanical Engineering student with a 4.0 GPA (Graduating in December 2027) specializing in robotics and autonomous systems. Experienced in control systems, CAD design, HVAC, engineering design, and simulation through research and engineering projects.

EDUCATION

Embry-Riddle Aeronautical University (ERAU)
Bachelor of Science: Mechanical Engineering
Area of Concentration: Robotics and Autonomous Systems

Daytona Beach, Florida
December 2027
GPA: 4.0

SKILLS

Computer Programming: MATLAB, Python, C++, Arduino, Microcontrollers
Modeling: SolidWorks, CATIA V5, Revit
Manufacturing & Tools: CNC Machining, Welding, Soldering, Precision Tolerancing, Technical Drawing, Excel, PowerPoint

PROFESSIONAL EXPERIENCES

HDR Inc., Pennington, New Jersey
Mechanical Engineer Intern

May 2026 - August 2026

- Contributed to HVAC design for the USCGA STEM West Project using Revit to develop ductwork routes, diffuser layouts, coordinate air distribution, and integrate mechanical systems within the building model.
- Used DuctSizer to size ductwork based on required airflow and design parameters, and selected Price square ceiling and laboratory exhaust diffusers for different space functions.
- Used Bluebeam Revu to review mechanical drawings, add design markups, compare revisions, and communicate coordination issues across project disciplines.

Hunt Library, Daytona Beach, Florida
Library Technician

April 2025 – Present

- Maintain accurate cataloging and inventory management for 10,000+ library resources using the Library of Congress classification system.
- Process and organize 100+ materials per shift, ensuring accurate shelving and system updates.
- Provide technical assistance to 30-50 students and faculty daily, resolving access system issues.

ENGINEERING PROJECTS

Yarkovsky-Kessler Mitigation Research - Undergraduate Researcher

Aug 2025 - Present

- Developed Python-based orbital simulation models to analyze Yarkovsky-induced perturbations affecting lunar debris.
- Modeled debris geometry, thermal dynamics, and orbital decay behavior in simulated space environments.
- Identified that irregularly shaped debris exhibits $\sim 3\times$ faster orbital decay than spherical bodies.

ERAU Manufacturing Lab - CNC Control Engineer

Jan 2026 - Present

- Redesigned a medical-grade mandrel centering guide in SolidWorks for a for B. Braun Medical's Class B clean room manufacturing environment.
- Fabricated precision prototype using CNC machining, welding, and soldering while maintaining tight dimensional tolerances.
- Evaluated anodized 6061 aluminum for fatigue resistance and cleanroom compatibility in medical device production.

The CATIA Final Project - Desk Lamp: Individual Assignment

Feb 2025 – Apr 2025

- Designed and modeled a multifunctional desk lamp in CATIA, integrating a drawer, phone cradle, and storage compartment.
- Produced fully dimensioned part and assembly drawings to industry standards with tight tolerances and manufacturable geometry.

The Halloween Animatronic Project: Microcontroller Programmer

Sep 2024 – Oct 2024

- Programmed Arduino-based motion-activated control with ultrasonic sensors.
- Designed and debugged automation logic to trigger synchronized body movements, creating an interactive animatronic system for public display.

HONORS AND AWARDS

Dean's List (Fall 2024- Present), Society of Women in Engineering Scholarship (2026), AFCEA STEM Scholarship (2026), Charles River All-Access Scholarship (2026), National STEM Excellence Scholarship (2025); ASME Scholarships (2025-2026), Women in STEM Scholarship (2024-2028), ERAU Excellence Award (2024-2028)